

# TM Forum Specification

## Promotion Management User Guide

**TMF671**

|                            |                                |
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# Introduction

The following document is the user guide of the REST API for Promotions management. It includes the model definition as well as all available operations.

This document describes a single managed resource called **Promotion**.

Promotions refer to additional discounts, vouchers, bonuses, or gifts provided to a party who meets predefined criteria. Promotions play a significant role in increasing product sales. They can be part of a larger campaign to raise awareness about a new product or a targeted objective.

This API is focused on the configuration and maintenance of this business object, not its execution.

The execution of promotions is the responsibility of the enterprise platform, which applies the discount (or gift, voucher, etc.) based on the information retrieved from the Promotion API.

Promotion API provides the following operations:

- Listing existing promotions based on filter criteria;
- Retrieving an existing promotion by identifier;
- Partially updating an existing promotion;
- Creating (configuring) a new promotion;
- Deleting an existing promotion;
- Notification events:
  - Promotion creation;
  - Promotion deletion;
  - Promotion attribute update;
  - Promotion state update.

## SAMPLE USE CASES

This section includes a set of main use cases that can be performed with the Promotion API. Additional use cases can be generated using the operations and resources defined in this specification.

### *Use Case 1: Promotion for prepaid customers*

|                  |                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User case        | The prepaid customer tops up their phone, qualifies for a particular promotion, and receives it.                                                                                                                                                                                                                                                |
| Actors           | Customer.                                                                                                                                                                                                                                                                                                                                       |
| Pre-conditions   | Qualified promotion configured in the catalog and released.                                                                                                                                                                                                                                                                                     |
| Expected outcome | Promotion benefits being provided to the customer.                                                                                                                                                                                                                                                                                              |
| Operation        | <b>GET /promotion</b>                                                                                                                                                                                                                                                                                                                           |
| Description      | <ol style="list-style-type: none"><li>1. Customer gets a recommendation notification about a promotion after doing a top-up of £25;</li><li>2. Customer checks its details:</li><li>3. "Top-up by £20 or more and get 5GB of data for only £5".</li><li>4. Customer appreciates the promotion and buys the 5GB of data top up add-on.</li></ol> |
| Note             | Purchase is handled by other API (TMF622).                                                                                                                                                                                                                                                                                                      |

### *Use Case 2: Create a new promotion*

|                  |                                                                                                                                                                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User case        | Creation of a new promotion related to customer retention.                                                                                                                                                                                                                                                                             |
| Actors           | Agent.                                                                                                                                                                                                                                                                                                                                 |
| Pre-conditions   | None.                                                                                                                                                                                                                                                                                                                                  |
| Expected outcome | New promotion configured.                                                                                                                                                                                                                                                                                                              |
| Operation        | <b>POST /promotion</b>                                                                                                                                                                                                                                                                                                                 |
| Description      | <ol style="list-style-type: none"><li>1. Agent receives a request to create a new promotion where customers receive 3 months of free Netflix subscription when they extend their contract by an additional 24 months;</li><li>2. Agent submits promotion details;</li><li>3. A new promotion with status "draft" is created.</li></ol> |

### *Use Case 3: Retire a promotion*

|                  |                                                                                                                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User case        | Retirement of a promotion.                                                                                                                                                                                          |
| Actors           | Agent.                                                                                                                                                                                                              |
| Pre-conditions   | Promotion previously released.<br>Promotion details are obsolete.                                                                                                                                                   |
| Expected outcome | Promotion status updated.                                                                                                                                                                                           |
| Operation        | <b>PATCH /promotion/id</b>                                                                                                                                                                                          |
| Description      | <ol style="list-style-type: none"><li>1. Agent receives a request to archive a promotion;</li><li>2. Agent submits a new status of the promotion;</li><li>3. Promotion status is changed to "retirement".</li></ol> |

*Use Case 4: Promotion for digital channels*

|                  |                                                                                                                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User case        | Promotion offering a 10% discount for new purchases through digital channels.                                                                                                                                                       |
| Actors           | Customer.                                                                                                                                                                                                                           |
| Pre-conditions   | Promotion configured in the catalog and released.                                                                                                                                                                                   |
| Expected outcome | Promotion presented to the customer.                                                                                                                                                                                                |
| Operation        | <b>GET /promotion</b>                                                                                                                                                                                                               |
| Description      | <ol style="list-style-type: none"><li>1. Customer logs into the digital portal;</li><li>2. Customer receives a recommendation notification about a 10% discount for new purchases;</li><li>3. Customer ignores promotion.</li></ol> |

## Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a `BillingAccount` and `SettlementAccount` inheriting properties from the abstract `Account` entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v3.0 Part 2 document.

The `@type` attribute provides a way to represent the actual class type of an entity. For example, within a list of `Account` instances some may be instances of `BillingAccount` where other could be instances of `SettlementAccount`. The `@type` gives this information. All resources and sub-resources of this API have a `@type` attributes that can be provided when this is useful.

The `@referredType` can be used within reference entities (like for instance an `AccountRef` object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the `@type`, when used, denotes the class type of the reference itself, such as `BillingAccountRef` or `SettlementAccountRef`, and not the class type of the referred object. However since reference classes are rarely sub-classed, `@type` is generally not useful in reference objects.

The `@schemaLocation` property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected *characteristics* of an entity.

The `@baseType` attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

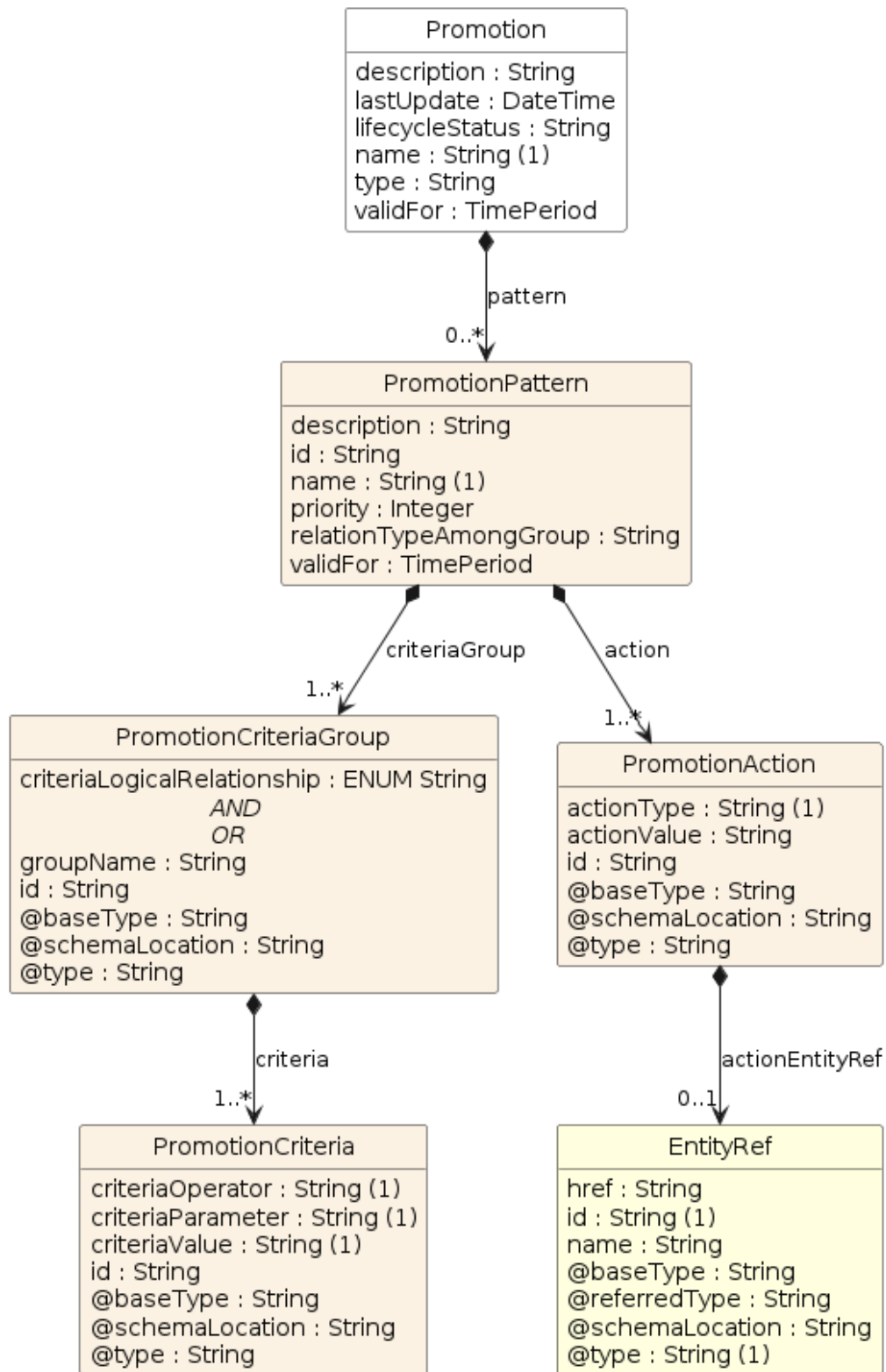
# RESOURCE MODEL

## Managed Entity and Task Resource Models

### Promotion resource

Promotion Resource is used to provide additional discounts, vouchers, bonuses, or gifts to customers who meet predefined criteria. By using promotions, the enterprise can attract users and encourage more consumption, especially continuous purchases. Normally, promotions are not considered a specific type of product or product offering. They are often applied when customers purchase product offerings that exceed a certain price or amount limit.

### Resource model



| Color | Description                     |
|-------|---------------------------------|
|       | Resource (entry point)          |
|       | Sub-resource                    |
|       | Sub-resource (reference entity) |

(1) : Mandatory property

Figure 1 - Promotion

**Field descriptions****Promotion fields**

|                 |                                                                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| description     | A String. Description of Promotion.                                                                                                                                           |
| lastUpdate      | A DateTime. Latest update date of Promotion.                                                                                                                                  |
| lifecycleStatus | A String. Status of Promotion, including draft/Test/WaitForApproval/Release/Suspend/Retirement.                                                                               |
| name            | A String. Name of Promotion.                                                                                                                                                  |
| pattern         | A PromotionPattern. Detailed pattern of the promotion. The pattern decides the conditions of promotion and the benefit of the promotion to be given to the eligible customer. |
| type            | A String. Type of promotion. The basic type is Award/Discount/Reduction. More types can be extended in future.                                                                |
| validFor        | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.                                                                        |

**PromotionPattern sub-resource fields**

|                        |                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| action                 | A PromotionAction. Action of the promotion. When the customer meets the conditions in the promotion pattern, the customer can be given the benefits in the action. |
| criteriaGroup          | A PromotionCriteriaGroup. Set of group criteria to promotion. Fulfilling these criteria, parties will receive benefits.                                            |
| description            | A String. Description.                                                                                                                                             |
| id                     | A String. Unique identifier.                                                                                                                                       |
| name                   | A String. Name.                                                                                                                                                    |
| priority               | An Integer. Priority. Smaller number means high.                                                                                                                   |
| relationTypeAmongGroup | A String. AND/OR, The logical relation type amongst the promotion criteria group.                                                                                  |
| validFor               | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.                                                             |

**PromotionCriteriaGroup sub-resource fields**

|                             |                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------|
| criteria                    | A PromotionCriteria. Set of criteria to be followed by all parties.                           |
| criteriaLogicalRelationship | A String. Logical relation followed by all criteria. ENUMERATED with values:<br>* AND<br>* OR |
| groupName                   | A String. Name of the group to be easily identified.                                          |
| id                          | A String. Unique Identifier.                                                                  |
| @baseType                   | A String. The base type for use in polymorphic collections.                                   |

|                 |                                                                            |
|-----------------|----------------------------------------------------------------------------|
| @schemaLocation | A String. A link to the schema describing a resource (for type extension). |
| @type           | A String. The class type of the actual resource (for type extension).      |

**PromotionAction sub-resource fields**

|                 |                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| actionEntityRef | An EntityRef. Entity reference schema to be use for all entityRef class.                                                                                                                                                                                                                                                                        |
| actionType      | A String. Action type can be one of the following: voice (minute), data (MB), data (GB), SMS, bonus point, physical gift, voucher, promotion code, an existing offering, currency.                                                                                                                                                              |
| actionValue     | A String. When the Promotion type is 1: Award, it means the amount or value of the awards decided by actionType, such as: Amount of gift, Amount of bonus, Value of discount; ? When the Promotion type is 2: Discount, it means the value of the discount; ? When the Promotion type is 3: reduction, it means the value of the reduced money. |
| id              | A String. Unique identifier.                                                                                                                                                                                                                                                                                                                    |
| @baseType       | A String. The base type for use in polymorphic collections.                                                                                                                                                                                                                                                                                     |
| @schemaLocation | A String. A link to the schema describing a resource (for type extension).                                                                                                                                                                                                                                                                      |
| @type           | A String. The class type of the actual resource (for type extension).                                                                                                                                                                                                                                                                           |

**PromotionCriteria sub-resource fields**

|                   |                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| criteriaOperator  | A String. Logic operator of this criteria instruction: '=' or '>' or '<' or '>=' or '□' or '<>'.                                                 |
| criteriaParameter | A String. The parameter (factor) of the criteria. The basic factors are abstracted from these data sources. There are several detail parameters. |
| criteriaValue     | A String. The value is filled for the comparison of the criteria.                                                                                |
| id                | A String. Unique identifier.                                                                                                                     |
| @baseType         | A String. The base type for use in polymorphic collections.                                                                                      |
| @schemaLocation   | A String. A link to the schema describing a resource (for type extension).                                                                       |
| @type             | A String. The class type of the actual resource (for type extension).                                                                            |

**EntityRef sub-resource fields**

|      |                                        |
|------|----------------------------------------|
| href | A String. Hyperlink reference.         |
| id   | A String. Unique identifier.           |
| name | A String. Name of the referred entity. |

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @referredType   | A String. The actual type of the target instance when needed for disambiguation.            |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String. When sub-classing, this defines the sub-class Extensible name.                    |

### Json representation sample(s)

We provide below a JSON representation as example of the 'Promotion' resource object.

```
{
  "id": "PREPAIDOFFER_5GB",
  "href": "https://host:port/tmf-api/promotion/v5/promotion/8474",
  "description": "Top-up by ?20 or more and get 5GB of data for only ?5",
  "lastUpdate": "2021-01-19T00:00:00.000Z",
  "lifecycleStatus": "release",
  "name": "More 5GB data when ?20 top-up",
  "pattern": [{
    "@type": "PromotionPattern",
    "id": "PATTERN_PREPAID_PRODUCT_OFFERING_5GB",
    "action": [{
      "@type": "PromotionAction",
      "id": "ACTION_PREPAID_PRODUCT_OFFERING_5GB",
      "actionEntityRef": {
        "@type": "EntityRef",
        "id": "PRODUCT_OFFERING_5GB",
        "href": "https://mycsp.com:8080/tmf-api/productCatalogManagement/v5/productOffering/PRODUCT_OFFERING_5GB",
        "name": "5GB data add-on"
      },
      "actionType": "3",
      "actionValue": "5"
    }],
    "criteriaGroup": [{
      "@type": "PromotionCriteriaGroup",
      "id": "GROUP_PREPAID_PRODUCT_OFFERING_5GB",
      "criteria": [{
        "@type": "PromotionCriteria",
        "id": "CRITERIA_PREPAID_PRODUCT_OFFERING_5GB",
        "criteriaOperator": ">=",
        "criteriaParameter": "5.3",
        "criteriaValue": "?20"
      }]
    }]
  }],
  "promotionType": "Reduction",
  "@type": "Promotion",
  "validFor": {
    "startDateTime": "2020-01-05T12:00:56.982Z",
    "endDateTime": "9999-12-31T23:59:59.999Z"
  }
}
```

## Notification Resource Models

4 notifications are defined for this API.

Notifications related to Promotion:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

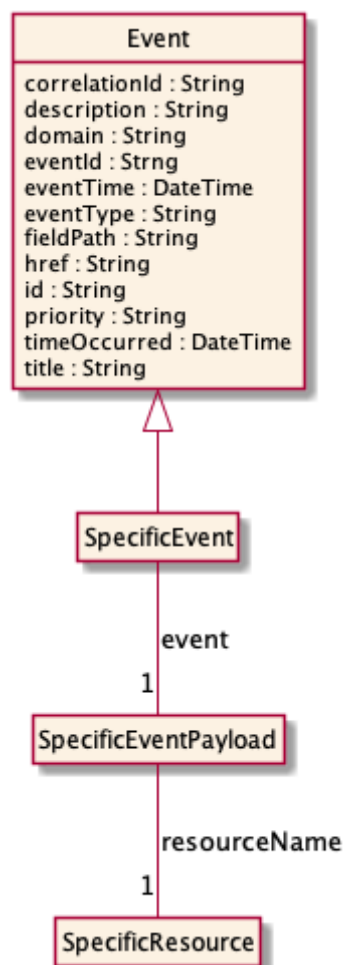


Figure 2 Notification Pattern

## Promotion

### Create Event

Message example for the PromotionCreateEvent event.

Content-Type: application/json

```
{
  "correlationId": "80c3041f-75f0",
  "description": "PromotionCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "4598-9729-305e4be5fb40",
  "eventTime": "2024-04-26T06:49:11.086Z",
  "eventType": "PromotionCreateEvent",
  "priority": "2",
  "timeOccurred": "2024-04-26T06:49:01.548Z",
  "title": "PromotionCreateEvent",
  "event": {
    "promotion": {
      "id": "PREPAIDOFFER_5GB",
      "href": "http://servername/promotion/PREPAIDOFFER_5GB",
      "description": "Top-up by ?20 or more and get 5GB of data for only ?5",
      "lastUpdate": "2021-01-19T00:00:00.000Z",
      "lifecycleStatus": "release",
      "name": "More 5GB data when ?20 top-up",
      "pattern": [
        {
          "@type": "PromotionPattern",
          "id": "PATTERN_PREPAID_PRODUCT_OFFERING_5GB",
          "action": [
            {
              "@type": "PromotionAction",
              "id": "ACTION_PREPAID_PRODUCT_OFFERING_5GB",
              "actionEntityRef": {
                "@type": "EntityRef",
                "id": "PRODUCT_OFFERING_5GB",
                "href": "https://mycsp.com:8080/tmf-api/productCatalogManagement/v5/productOffering/PRODUCT_OFFERING_5GB",
                "name": "5GB data add-on"
              },
              "actionType": "3",
              "actionValue": "5"
            }
          ],
          "criteriaGroup": [
            {
              "@type": "PromotionCriteriaGroup",
              "id": "GROUP_PREPAID_PRODUCT_OFFERING_5GB",
              "criteria": [
                {
                  "@type": "PromotionCriteria",
                  "id": "CRITERIA_PREPAID_PRODUCT_OFFERING_5GB",
                  "criteriaOperator": ">=",
                  "criteriaParameter": "5.3",
                  "criteriaValue": "?20"
                }
              ]
            }
          ]
        }
      ]
    }
  }
}
```

```

    ],
    "@type": "Promotion",
    "promotionType": "Reduction",
    "validFor": {
        "startDateTime": "2020-01-05T12:00:56.982Z",
        "endDateTime": "9999-12-31T23:59:59.999Z"
    }
  },
  "reportingSystem": {
    "id": "634",
    "name": "APP-849",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "657",
    "name": "APP-400",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PromotionCreateEvent"
}

```

## Delete Event

Message example for the PromotionDeleteEvent event.

Content-Type: application/json

```

{
  "correlationId": "7651dffe-da1a",
  "description": "PromotionDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "48ab-8286-421f5ebc94d5",
  "eventTime": "2024-04-26T06:49:11.087Z",
  "eventType": "PromotionDeleteEvent",
  "priority": "1",
  "timeOccurred": "2024-04-26T06:49:09.867Z",
  "title": "PromotionDeleteEvent",
  "event": {
    "promotion": {
      "id": "PREPAIDOFFER_5GB",
      "href": "http://servername/promotion/PREPAIDOFFER_5GB",
      "@type": "Promotion"
    }
  },
  "reportingSystem": {
    "id": "634",
    "name": "APP-849",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "657",
    "name": "APP-400",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
}

```

```

"@baseType": "Event",
"@type": "PromotionDeleteEvent"
}

```

## Attribute Value Change Event

Message example for the PromotionAttributeValueChangeEvent event.

```

Content-Type: application/json

{
  "correlationId": "d71f301f-36d7",
  "description": "PromotionAttributeValueChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "46be-b942-f9e7f72886ae",
  "eventTime": "2024-04-26T06:49:11.087Z",
  "eventType": "PromotionAttributeValueChangeEvent",
  "priority": "1",
  "timeOccurred": "2024-04-26T06:49:04.322Z",
  "title": "PromotionAttributeValueChangeEvent",
  "event": {
    "promotion": {
      "href": "http://servername/promotion/PREPAIDOFFER_5GB",
      "id": "PREPAIDOFFER_5GB",
      "validFor": {
        "startDateTime": "2020-01-05T12:00:56.982Z",
        "endDateTime": "9999-12-31T23:59:59.999Z"
      },
      "@type": "Promotion"
    }
  },
  "reportingSystem": {
    "id": "634",
    "name": "APP-849",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "657",
    "name": "APP-400",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PromotionAttributeValueChangeEvent"
}

```

## State Change Event

Message example for the PromotionStateChangeEvent event with ?fields=state property only.

```

Content-Type: application/json

{
  "correlationId": "54a25dc9-51da",
  "description": "PromotionStateChangeEvent illustration",

```

```
{
  "domain": "Commercial",
  "eventId": "4529-a42e-715cb69ecfaf",
  "eventTime": "2024-04-26T06:49:11.087Z",
  "eventType": "PromotionStateChangeEvent",
  "priority": "3",
  "timeOccured": "2024-04-26T06:49:04.495Z",
  "title": "PromotionStateChangeEvent",
  "event": {
    "promotion": {
      "id": "PREPAIDOFFER_5GB",
      "href": "http://servername/promotion/PREPAIDOFFER_5GB",
      "lifecycleStatus": "retirement",
      "@type": "Promotion"
    }
  },
  "reportingSystem": {
    "id": "634",
    "name": "APP-849",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "657",
    "name": "APP-400",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PromotionStateChangeEvent"
}
```

# API OPERATIONS

Remember the following Uniform Contract:

| Operation on Entities          | Uniform API Operation | Description                                                                     |
|--------------------------------|-----------------------|---------------------------------------------------------------------------------|
| Query Entities                 | GET Resource          | GET must be used to retrieve a representation of a resource.                    |
| Create Entity                  | POST Resource         | POST must be used to create a new resource                                      |
| Partial Update of an Entity    | PATCH Resource        | PATCH must be used to partially update a resource                               |
| Complete Update of an Entity   | PUT Resource          | PUT must be used to completely update a resource identified by its resource URI |
| Remove an Entity               | DELETE Resource       | DELETE must be used to remove a resource                                        |
| Execute an Action on an Entity | POST on TASK Resource | POST must be used to execute Task Resources                                     |
| Other Request Methods          | POST on TASK Resource | GET and POST must not be used to tunnel other request methods.                  |

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

## Operations on Promotion

### List or find Promotion objects

**GET** /promotion?fields=...

#### Description

This operation list Promotion entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

#### Usage samples

Here is an example of a request for retrieving multiple Promotions.

#### Request

```
GET /promotion
Content-Type: application/json
```

#### Response

200

```
[
  {
    "id": "CUSTBIRTHDAY",
    "href": "https://host:port/tmf-api/promotion/v5/promotion/CUSTBIRTHDAY",
    "description": "It's your birthday! 50 loyalty points will be added to your account",
    "lastUpdate": "2000-01-19T00:00:00.000Z",
    "lifecycleStatus": "retirement",
    "name": "50 loyalty points just to you",
    "pattern": [{
      "@type": "PromotionPattern",
      "id": "PATTERN_50_LOYALTY_POINTS",
      "action": [{
        "@type": "PromotionAction",
        "id": "ACTION_50_LOYALTY_POINTS",
        "actionType": "10",
        "actionValue": "50 loyalty points"
      }],
      "criteriaGroup": [{
        "@type": "PromotionCriteriaGroup",
        "id": "GROUP_50_LOYALTY_POINTS",
        "criteria": [{
          "@type": "PromotionCriteria",
          "id": "CRITERIA_50_LOYALTY_POINTS",
          "criteriaOperator": "=",
          "criteriaParameter": "1.3",
          "criteriaValue": "02-09"
        }]
      }]
    }],
    "promotionType": "Award",
    "validFor": {
      "startDateTime": "2020-01-05T12:00:56.982Z",
      "endDateTime": "9999-12-31T23:59:59.999Z"
    },
    "@type": "Promotion"
  }, {
    "id": "RETENTIONNETFLIX3M",
    "href": "https://host:port/tmf-api/promotion/v5/promotion/RETENTIONNETFLIX3M",
    "description": "Retention for more 24 months gives free Netflix for 3months",
    "lastUpdate": "2021-02-10T00:00:00.000Z",
    "lifecycleStatus": "draft",
    "name": "3months of free Netflix when retention for more 24 months",
    "pattern": [{
      "@type": "PromotionPattern",
      "action": [{
        "@type": "PromotionAction",
        "actionEntityRef": {
          "@type": "EntityRef",
          "id": "POSTPAID_PRODUCT_OFFERING_NETFLIX",
          "href": "https://mycsp.com:8080/tmf-api/productCatalogManagement/v5/productOffering/PRODUCT_OFFERING_NETFLIX",
          "name": "Netflix 3MFree"
        },
        "actionType": "9"
      }],
      "criteriaGroup": [{
        "@type": "PromotionCriteriaGroup",
        "criteria": [{
          "@type": "PromotionCriteria",
```

```

        "criteriaOperator": "=",
        "criteriaParameter": "3.2",
        "criteriaValue": "24"
      }, {
        "@type": "PromotionCriteria",
        "criteriaOperator": "=",
        "criteriaParameter": "3.3",
        "criteriaValue": "months"
      }
    ]
  },
  "promotionType": "Award",
  "validFor": {
    "startDateTime": "2020-01-05T12:00:56.982Z",
    "endDateTime": "9999-12-31T23:59:59.999Z"
  },
  "@type": "Promotion"
}, {
  "id": "PREPAIDOFFER_5GB",
  "href": "https://host:port/tmf-api/promotion/v5/promotion/PREPAIDOFFER_5GB",
  "description": "Top-up by ?20 or more and get 5GB of data for only ?5",
  "lastUpdate": "2021-02-10T00:00:00.000Z",
  "lifecycleStatus": "release",
  "name": "More 5GB data when 20euros top-up",
  "pattern": [{
    "@type": "PromotionPattern",
    "id": "PATTERN_PREPAID_PRODUCT_OFFERING_5GB",
    "action": [{
      "@type": "PromotionAction",
      "id": "ACTION_PREPAID_PRODUCT_OFFERING_5GB",
      "actionEntityId": {
        "@type": "EntityRef",
        "id": "PRODUCT_OFFERING_5GB",
        "href": "https://mycsp.com:8080/tmf-api/productCatalogManagement/v5/productOffering/PRODUCT_OFFERING_5GB",
        "name": "5GB data add-on"
      },
      "actionType": "3",
      "actionValue": "5"
    }
  ]},
  "criteriaGroup": [{
    "@type": "PromotionCriteriaGroup",
    "id": "GROUP_PREPAID_PRODUCT_OFFERING_5GB",
    "criteria": [{
      "@type": "PromotionCriteria",
      "id": "CRITERIA_PREPAID_PRODUCT_OFFERING_5GB",
      "criteriaOperator": ">=",
      "criteriaParameter": "5.1",
      "criteriaValue": "?20"
    }
  ]
}
],
  "promotionType": "Reduction",
  "validFor": {
    "startDateTime": "2020-01-05T12:00:56.982Z",
    "endDateTime": "9999-12-31T23:59:59.999Z"
  },
  "@type": "Promotion"
}]

```

## Retrieves a Promotion by ID

**GET /promotion/{id}?fields=...**

### Description

This operation retrieves a Promotion entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

### Usage samples

Here is an example of a request for retrieving a specific Promotion.

### Request

```
GET /promotion/RETENTIONNETFLIX3M
Content-Type: application/json
```

### Response

```
200

{
  "id": "RETENTIONNETFLIX3M",
  "href": "https://host:port/tmf-api/promotion/v5/promotion/RETENTIONNETFLIX3M",
  "description": "Retention for more 24 months gives free Netflix for 3 months",
  "lastUpdate": "2021-02-10T00:00:00.000Z",
  "lifecycleStatus": "draft",
  "name": "3 months of free Netflix when retention for more 24 months",
  "pattern": [{
    "@type": "PromotionPattern",
    "action": [{
      "@type": "PromotionAction",
      "actionEntityRef": {
        "@type": "EntityRef",
        "id": "POSTPAID_PRODUCT_OFFERING_NETFLIX",
        "href": "https://mycsp.com:8080/tmf-api/productCatalogManagement/v5/productOffering/PRODUCT_OFFERING_NETFLIX",
        "name": "Netflix 3MFree"
      },
      "actionType": "9"
    }],
    "criteriaGroup": [{
      "@type": "PromotionCriteriaGroup",
      "criteria": [{
        "@type": "PromotionCriteria",
        "criteriaOperator": "=",
        "criteriaParameter": "3.2",
        "criteriaValue": "24"
      }, {
        "@type": "PromotionCriteria",
        "criteriaOperator": "=",
        "criteriaParameter": "3.3",
        "criteriaValue": "months"
      }
    ]
  }]
}
```

```

    },
    "promotionType": "Award",
    "validFor": {
      "startDateTime": "2020-01-05T12:00:56.982Z",
      "endDateTime": "9999-12-31T23:59:59.999Z"
    },
    },
    "@type": "Promotion"
  }
}

```

## Creates a Promotion

### POST /promotion?fields=...

#### Description

This operation creates a Promotion entity.

#### Mandatory Attributes

| Mandatory Attributes | Rule |
|----------------------|------|
| name                 |      |

#### Usage samples

Here is an example of a request for creating a Promotion.

#### Request

```

POST /promotion
Content-Type: application/json

{
  "description": "Retention for more 24 months gives free Netflix for 3 months",
  "lastUpdate": "2021-01-19T00:00:00.000Z",
  "lifecycleStatus": "draft",
  "name": "3 months of free Netflix when retention for more 24 months",
  "pattern": [{
    "@type": "PromotionPattern",
    "action": [{
      "actionEntityRef": {
        "@type": "EntityRef",
        "id": "POSTPAID_PRODUCT_OFFERING_NETFLIX",
        "href": "https://mycsp.com:8080/tmf-api/productCatalogManagement/v5/productOffering/PRODUCT_OFFERING_NETFLIX",
        "name": "Netflix 3MFree"
      },
      "actionType": "9"
    }],
    "criteriaGroup": [{
      "@type": "PromotionCriteriaGroup",
      "criteria": [{
        "@type": "PromotionCriteria",
        "criteriaOperator": "=",
        "criteriaParameter": "3.2",
        "criteriaValue": "24"
      }],
    }],
  }],
}

```

```

        "@type": "PromotionCriteria",
        "criteriaOperator": "=",
        "criteriaParameter": "3.3",
        "criteriaValue": "months"
    }
  ]
},
"promotionType": "Award",
"@type": "Promotion",
"validFor": {
  "startDateTime": "2020-01-05T12:00:56.982Z",
  "endDateTime": "9999-12-31T23:59:59.999Z"
}
}

```

## Response

```

201

{
  "id": "RETENTIONNETFLIX3M",
  "href": "https://host:port/tmf-api/promotion/v5/promotion/RETENTIONNETFLIX3M",
  "description": "Retention for more 24 months gives free Netflix for 3 months",
  "lastUpdate": "2021-01-19T00:00:00.000Z",
  "lifecycleStatus": "draft",
  "name": "3 months of free Netflix when retention for more 24 months",
  "pattern": [{
    "@type": "PromotionPattern",
    "id": "POSTPAID_PRODUCT_OFFERING_NETFLIX",
    "action": [{
      "actionEntityRef": {
        "@type": "EntityRef",
        "id": "POSTPAID_PRODUCT_OFFERING_NETFLIX",
        "href": "https://mycsp.com:8080/tmf-api/productCatalogManagement/v5/productOffering/PRODUCT_OFFERING_NETFLIX",
        "name": "Netflix 3MFree"
      },
      "actionType": "9"
    }],
    "criteriaGroup": [{
      "@type": "PromotionCriteriaGroup",
      "id": "POSTPAID_PRODUCT_OFFERING_NETFLIX",
      "criteria": [{
        "@type": "PromotionCriteria",
        "id": "POSTPAID_PRODUCT_OFFERING_NETFLIX",
        "criteriaOperator": "=",
        "criteriaParameter": "3.2",
        "criteriaValue": "24"
      }, {
        "@type": "PromotionCriteria",
        "id": "POSTPAID_PRODUCT_OFFERING_NETFLIX",
        "criteriaOperator": "=",
        "criteriaParameter": "3.3",
        "criteriaValue": "months"
      }
    ]
  }],
  "promotionType": "Award",
  "@type": "Promotion",
  "validFor": {

```

```

    "startDateTime": "2020-01-05T12:00:56.982Z",
    "endDateTime": "9999-12-31T23:59:59.999Z"
  }
}

```

## Updates partially a Promotion

**PATCH** /promotion/{id}?fields=...

### Description

This operation allows partial updates of a Promotion entity. Support of json/merge (<https://tools.ietf.org/html/rfc7396>) is mandatory, support of json/patch (<http://tools.ietf.org/html/rfc5789>) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

### Patchable and Non Patchable Attributes

| Non Patchable Attributes | Rule |
|--------------------------|------|
|                          |      |
| Patchable Attributes     | Rule |
| description              |      |
| lastUpdate               |      |
| lifecycleStatus          |      |
| name                     |      |
| pattern                  |      |
| type                     |      |
| validFor                 |      |

### Usage samples

Here is an example of a request for partially updating a Promotion.

### Request

```

PATCH /promotion/CUSTBIRTHDAY
Content-Type: application/merge-patch+json

{
  "lifecycleStatus": "retirement",
  "@type": "Promotion"
}

```

### Response

```

200

{
  "id": "CUSTBIRTHDAY",
  "href": "https://host:port/tmf-api/promotion/v5/promotion/CUSTBIRTHDAY",
  "description": "It's your birthday! 50 loyalty points will be added to your account",
  "lastUpdate": "2000-01-19T00:00:00.000Z",
  "lifecycleStatus": "retirement",
  "name": "50 loyalty points just to you",
  "pattern": [{
    "@type": "PromotionPattern",
    "id": "PATTERN_50_LOYALTY_POINTS",
    "action": [{
      "@type": "PromotionAction",
      "id": "ACTION_50_LOYALTY_POINTS",
      "actionType": "10",
      "actionValue": "50 loyalty points"
    }],
    "criteriaGroup": [{
      "@type": "PromotionCriteriaGroup",
      "id": "GROUP_50_LOYALTY_POINTS",
      "criteria": [{
        "@type": "PromotionCriteria",
        "id": "CRITERIA_50_LOYALTY_POINTS",
        "criteriaOperator": "=",
        "criteriaParameter": "1.3",
        "criteriaValue": "03-09"
      }]
    }]
  }],
  "promotionType": "Award",
  "validFor": {
    "startDateTime": "2020-01-05T12:00:56.982Z",
    "endDateTime": "9999-12-31T23:59:59.999Z"
  },
  "@type": "Promotion"
}

```

Here is an example of a request for partially updating a Promotion.

### Request

```

PATCH /promotion/CUSTBIRTHDAY
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/lifecycleStatus",
    "value": "retirement"
  }
]

```

### Response

```

200

{
  "id": "CUSTBIRTHDAY",
  "href": "https://host:port/tmf-api/promotion/v5/promotion/CUSTBIRTHDAY",
  "description": "It's your birthday! 50 loyalty points will be added to your account",
  "lastUpdate": "2000-01-19T00:00:00.000Z",
  "lifecycleStatus": "retirement",
  "name": "50 loyalty points just to you",
  "pattern": [{
    "@type": "PromotionPattern",
    "id": "PATTERN_50_LOYALTY POINTS",
    "action": [{
      "@type": "PromotionAction",
      "id": "ACTION_50_LOYALTY POINTS",
      "actionType": "10",
      "actionValue": "50 loyalty points"
    }],
    "criteriaGroup": [{
      "@type": "PromotionCriteriaGroup",
      "id": "GROUP_50_LOYALTY POINTS",
      "criteria": [{
        "@type": "PromotionCriteria",
        "id": "CRITERIA_50_LOYALTY POINTS",
        "criteriaOperator": "=",
        "criteriaParameter": "1.3",
        "criteriaValue": "03-09"
      }]
    }]
  }],
  "promotionType": "Award",
  "validFor": {
    "startDateTime": "2020-01-05T12:00:56.982Z",
    "endDateTime": "9999-12-31T23:59:59.999Z"
  },
  "@type": "Promotion"
}

```

Here is an example of a request for partially updating a Promotion.

### Request

```

PATCH /promotion/CUSTBIRTHDAY
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/criteriaGroup/criteria/criteriaValue?id=CRITERIA_50_LOYALTY POINTS",
    "value": "03-09"
  }
]

```

### Response

200

```
{
  "id": "CUSTBIRTHDAY",
  "href": "https://host:port/tmf-api/promotion/v5/promotion/CUSTBIRTHDAY",
  "description": "It's your birthday! 50 loyalty points will be added to your
account",
  "lastUpdate": "2000-01-19T00:00:00.000Z",
  "lifecycleStatus": "retirement",
  "name": "50 loyalty points just to you",
  "pattern": [{
    "@type": "PromotionPattern",
    "id": "PATTERN_50_LOYALTY POINTS",
    "action": [{
      "@type": "PromotionAction",
      "id": "ACTION_50_LOYALTY POINTS",
      "actionType": "10",
      "actionValue": "50 loyalty points"
    }],
    "criteriaGroup": [{
      "@type": "PromotionCriteriaGroup",
      "id": "GROUP_50_LOYALTY POINTS",
      "criteria": [{
        "@type": "PromotionCriteria",
        "id": "CRITERIA_50_LOYALTY POINTS",
        "criteriaOperator": "=",
        "criteriaParameter": "1.3",
        "criteriaValue": "03-09"
      }]
    }]
  }],
  "promotionType": "Award",
  "validFor": {
    "startDateTime": "2020-01-05T12:00:56.982Z",
    "endDateTime": "9999-12-31T23:59:59.999Z"
  },
  "@type": "Promotion"
}
```

## Deletes a Promotion

**DELETE** /promotion/{id}

### Description

This operation deletes a Promotion entity.

### Usage samples

Here's an example of a request for deleting a Promotion.

### Request

```
DELETE /promotion/PREPAIDOFFER_5GB
Content-Type: application/json
{
  "id": "PREPAIDOFFER_5GB",
```

```
"href": "https://host:port/tmf-api/promotion/v5/promotion/PREPAIDOFFER_5GB",  
"@type": "Promotion"  
}
```

## Response

```
204  
{  
  "id": "PREPAIDOFFER_5GB",  
  "href": "https://host:port/tmf-api/promotion/v5/promotion/PREPAIDOFFER_5GB",  
  "@type": "Promotion"  
}
```

## API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

### Register listener

#### POST /hub

##### Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

##### Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

##### Usage Samples

Here's an example of a request for registering a listener.

##### Request

```
POST /api/hub

Accept: application/json

{
  "callback": "http://in.listener.com"
}
```

##### Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
  "id": "42",
  "callback": "http://in.listener.com",
  "query": ""
}
```

## Unregister listener

**DELETE /hub/{id}**

### Description

Clears the communication endpoint address that was set by creating the Hub..

### Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

### Usage Samples

Here's an example of a request for un-registering a listener.

### Request

```
DELETE /api/hub/42
Accept: application/json
```

### Response

```
204
```

## Publish Event to listener

**POST /client/listener**

### Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

### Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

### Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

### Request

```
POST /client/listener

Accept: application/json

{
  "event": {
    EVENT BODY
  },
  "eventType": "EVENT_TYPE"
}
```

## Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

# Acknowledgements

## Release History

| Release Number | Date       | Release led by:                                                                                                                        | Description                     |
|----------------|------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Release 1.0.0  | 04/15/2017 | Pierre Gauthier<br>TM Forum<br><a href="mailto:pgauthier@tmforum.org">pgauthier@tmforum.org</a><br><br>Mariano Belaunde<br>Orange Labs | First Release of the Document.  |
| Release 3.0.0  | 11/06/2018 | Mariano Belaunde<br>Orange Labs                                                                                                        | Alignment with Guidelines 3.0   |
| Release 4.0.0  | 23/06/2021 | Lara Silva<br>Vodafone                                                                                                                 | Alignment with Guidelines 4.0.1 |
| Release 5.0.0  | 03/05/2024 | Paul (Pavlo) Limarenko<br>Salesforce                                                                                                   | Alignment with Guidelines 5.0.0 |

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